W-G-690C
July 23, 1985
SUPERSEDING
W-G-690B
February 28, 1975

FEDERAL SPECIFICATION

GRINDING KITS, VALVE SEAT, ELECTRIC

This specification is approved by the Assistant Administrator, Office of Federal Supply and Services, General Services Administration, for the use of all Federal agencies.

1 SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers grinding kits to be used in the grinding of intake and exhaust valve seats located in engine cylinder blocks, cylinder heads, and also removable cylinders of air-cooled engines.

1.2 Classification.

1.2.1 Types and sizes. The grinding kits covered by this specification shall be of the following types and sizes as specified (see 6.2).

Type I - For grinding valve seats located in engine cylinder block or head.

Size 1 - Valve range 1 inch (2.5 cm.) to 3 inches (7.6 cm.).
Size 2 - Valve range 3 inches (7.6 cm.) to 4 inches (10.2 cm.)
Size 3 - Valve range 4 inches (10.2 cm.) to 6 inches (15.2 cm.).

Type II - For grinding valve seats located in removable engine cylinders.

Size 1 - Valve range 1 inch (2.5 cm.) to 3 inches (7.6 cm.).

2 APPLICABLE DOCUMENTS

2.1 Government publications. The issues of the following documents, in effect on date of invitation for bids or solicitation for offers, form a part of this specification to the extent specified herein.

FSC 4910

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Federal Specifications:

J-C-175 - Cable Assembly, Power, Electrical, For 125-Volt, 250-Volt, and 125/250-Volt, 50-60 Hz, Equipment
W-C-596/11 - Connector, Receptacle, Electrical, General Purpose, Single, Grounding, 2 Pole, 3 Wire, 15 Ampere, 125 Volts
CC-M-1807 - Motor, Alternating Current, Fractional and Integral Horsepower (500 HP and Smaller)
PPP-P-40 - Packaging and Packing of Hand Tools

Federal Standards:

FED-STD-H28 - Screw-thread Standards for Federal Services
FED-STD-123 - Marking for Shipment (Civil Agencies)

(Activities outside the Federal Government may obtain copies of Federal specifications, standards, and commercial item descriptions as outlined under General Information in the Index of Federal Specifications, Standards and Commercial Item Descriptions. The Index, which includes cumulative bimonthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, US Government Printing Office, Washington DC 20402.

(Single copies of this specification, and other Federal specifications and commercial item descriptions required by activities outside the Federal Government for bidding purposes are available without charge from General Services Administration Business Service Centers in Boston, MA; New York, NY; Philadelphia, PA; Washington, DC; Atlanta, GA; Chicago, IL; Kansas City, MO; Fort Worth, TX; Houston, TX; Denver, CO; San Francisco, CA; Los Angeles, CA; and Seattle, WA.

(Federal Government activities may obtain copies of Federal standardization documents and the Index of Federal Specifications, Standards and Commercial Item Descriptions from established distribution points in their agencies.)

Military Specifications:

MIL-P-116 - Preservation, Methods of
MIL-T-152 - Treatment, Moisture- and Fungus-Resistant, of Communications, Electronic, and Associated Electrical Equipment
MIL-V-173 - Varnish, Moisture- and Fungus-Resistant (For Treatment of Communications, Electronic, and Associated Equipment)

Military Standards:

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129 - Marking for Shipment and Storage
MIL-STD-130 - Identification Marking of US Military Property
MIL-STD-889 - Dissimilar Metals
MIL-STD-1337 - General Suppression System Design Requirements for Portable Electric Hand Tools

(Copies of Military specifications and standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

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2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on date of invitation for bids or request for proposal shall apply.

American Gear Manufacturers Association (AGMA):

AGMA 360.02 - Manual for Machine Tool Gearing
AGMA 390.03 - Gear Handbook, Volume 1, Gear Classification, Materials and Measuring Methods for Unassembled Gears

(Application for copies should be addressed to American Gear Manufacturers Association, 1901 North Fort Myer Drive, Arlington, Virginia 22209.)

American National Standards Institute, Inc. (ANSI):

ANSI B7.1 - Safety requirements for the Use, Care, and Protection of Abrasive Wheels
ANSI B74.13 - Markings for Identifying Grinding Wheels and Other Bonded Abrasives

(Application for copies should be addressed to the Amercian National Standards Institute Inc., 1430 Broadway, New York, NY 10018)

American Society for Testing and Materials (ASTM):

ASTM E380-84 - Standard for Metric Practice
ASTM D3951-82- Standard Practice for Commercial Packaging

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

International Organization for Standardization (ISO):

ISO 1122/1 - Glossary of Gears Terms - Part 1: Geometrical Definitions
ISO 54 - Cylindrical Gears for General Engineering and for Heavy Engineering - Modules and Diametral Pitches

(Application for copies should be addressed to the American National Standards Institute, Inc., 1430 Broadway, New York, NY 10018.)

National Fire Protection Association (NFPA):

NFPA 70 - National Electrical Code

(Application for copies should be addressed to the National Fire Protection Association, Batterymarch Park, Quincy, MA 02269.)

Underwriters' Laboratories (UL), Inc., Standard for Safety:

UL 45 - Portable Electric Tools
UL 73 - Motor-Operated Appliances

(Application for copies should be addressed to the Underwriters' Laboratories, Inc., 333 Pfingsten Road, Northbrook, IL 60612.)

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(Industry association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

3 REQUIREMENTS

3.1 First article. When specified (see 6.2), the contractor shall furnish one or more grinding kits for first article inspection. The first article may be either a preproduction model(s) or an initial production item(s) which conforms to the requirements of this specification. In either case, the approved first article and the production items shall be identical and in accordance with the terms of the contract. Approval of the first article shall not relieve the contractor of the responsibility to furnish equipment in accordance with the requirements of this specification.

3.2 Design. The grinding kits shall be new and shall include, as a minimum, the following components: electric-powered driver, grinding wheels, wheel holders, pilots, wheel dresser and stand, and valve-seat indicator; all components shall be enclosed in a carrying case with handles.

3.2.1 Maintainability. Design and assembly of the grinding kits and the placement of adjusting features shall insure that maintenance of operational capability will be reduced to the simplest form of procedures requiring a minimum of skill. Instructions for disassembly, repair, assembly, and adjustment shall be clear, concise, and definitive in application.

3.3 Materials. Materials not specifically designated herein or in the contract shall be of a quality commensurate with commercial practice within the grinding kit industry, shall be suitable for the intended purpose in the design of the end item, and shall meet all requirements specified herein. Materials shall be free from defects which would adversely affect the performance or maintainability of the individual components or the overall assembly. It shall not be permissible to use reclaimed parts as is, or rebuilt from scrap or other used equipment. When dissimilar metals are used in contact with each other, suitable protection against galvanic corrosion shall be applied in accordance with MIL-STD-889.

3.4 Construction. The grinding kits shall be complete so that when placed in use they shall be capable of performing the operations specified herein. Reclaimed parts shall not be used in construction of the grinding kits.

3.4.1 Castings and forgings. All castings and forgings shall be free from defects, scale, and mismatching. In no event shall such processes as welding, peening, plugging, or filling with solders or metallic pastes be used for reclaiming any defective castings or forgings.

3.4.2 Welding, brazing, and soldering. Welding, brazing, and soldering shall be of a quality which shall sustain all requirements of the welded, brazed, or soldered parts. These operations shall not be employed as repair measures for defective parts.

3.4.3 Fastening devices. All screws, pins, bolts, and similar parts shall be installed with means for preventing loss of tightness and adjustment. All such parts when subject to removal or adjustment shall not be swaged, peened, staked, or otherwise permanently deformed.

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3.4.4 Electrical connections. Connections of conductors and terminal parts shall be of the screw, pressure, or solder type. When soldered connections are used, the conductors and terminal parts shall be mechanically secured to insure electrical continuity before soldering. Rosin base fluxes only shall be used in soldering operations. Connections to screw-type terminals shall be mechanically secured with a means to prevent loss of tightness.

3.4.5 Threads. Threads shall conform to FED-STD-H28 and the applicable Detailed Standard section referenced therein.

3.4.6 Gears. All gear and pinion designs selected for use in the machine or its components shall be manufactured to either English (USA) or standard inch diametral pitch per AGMA 390.03, AGMA 360.02 or Metric (SI) module ratio per the International Organization for Standardization (ISO), ISO 1122/1, ISO 54 and ASTM E380.

3.4.7 Lubrication. All bearings (except sealed-for-life type), mating gears, and sliding parts shall be provided with standard means for lubrication. Oil holes, grease fittings and filler caps shall be accessible without requiring disassembly of functional parts. Lubrication reservoirs, if applicable, shall have means for checking levels.

3.4.8 Safety devices. The grinding wheels in the grinding kits shall comply with the applicable requirements of ANSI B7.1 and B74.13. Parts which are hazardous to the operator shall be guarded. Protection against electric shock shall be provided. When specified (see 6.2), special safety devices shall be provided as required by the procuring activity.

3.4.9 Interchangeability. All parts shall be manufactured to definite standards, clearances, and tolerances in order that any such parts of a particular type or model may be replaced, interchanged, and adjusted without modification of the replacement parts or any other parts of the unit. When practical, all parts shall be permanently and legibly marked in accordance with MIL-STD-130.

3.4.10 Surfaces. All castings, forgings, molded parts, stampings, and welded parts shall be cleaned and free of sand, dirt, fins, sprues, scale, flux, and other harmful or extraneous materials. External surfaces shall be smoothed and all edges shall be rounded or beveled.

3.5 Performance and product characteristics.

3.5.1 Functional and capacity requirements. The functional and capacity requirements for each type and size grinding kit shall be as follows.

3.5.1.1 Type I. The type I grinding kit shall be designed for grinding valve seats located in the engine cylinder block (with the engine in its applied position), and valve seats in engine cylinder heads (see Table I).

3.5.1.2 Type II. The type II grinding kit shall be designed to accommodate valve seats per Figure 1 in removable air-cooled engine cylinders (see Table I).

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3.5.2 Grinding accuracy. The grinding kits shall be capable of reconditioning (finish grinding) valve seats to be concentric with the center line of the valve guide hole to within 0.002 inch total indicator reading (tir). The surface roughness of the valve seat finished ground surface shall not exceed 32 micro-inches arithmetical average.

3.5.3 Vibration. The carrying case, complete with all components, shall withstand without damage simple harmonic motion at a frequency range of 5 to 35 hertz (Hz) for the type I kit and 5 to 55 Hz for the type II kit traversed once per minute at an amplitude of 1/32 inch (1/16 inch total excursion) through each of its three major axes for a period of 20 minutes in each axis.

3.5.4 Shock. The grinding kit with all components thereof placed in the carrying case shall withstand without damage the shock resulting from being dropped 3 times, on the bottom surface of the carrying case, at a vertical distance of 8 inches to a concrete surface.

3.6 Details of components. The grinding kit shall consist of, but shall not be limited to, the following principal component parts.

3.6.1 Driver. The grinder driver shall be a hand-held electric-powered rotary tool for driving grinding wheels.

3.6.1.1 Spindle drive. The spindle drive shall transmit power from the motor to the grinding wheels to perform the valve seat grinding requirements specified in Table I. If the wheel holder is not an integral part of the spindle, the drive end of the spindle drive shall incorporate provisions for connecting the grinding wheel holder specified in 3.6.3. The connection shall provide a positive fitting between the wheel holder and the spindle without slippage, looseness, and detrimental vibration during operation of the driver. The drive spindle for the type I grinding kit shall effect an included angle of 105 degrees to 150 degrees with the center line of the drive spindle and the center line of the armature shaft. If a straight drive unit is furnished, an angle attachment shall be provided. The drive spindle for the type II grinding kit shall be at an angle to accommodate grinding seats per Figure 1.

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TABLE I - CAPACITY

Type I				Type II
Capacities				
Size	1	2	3	1
Valve seat outside diameter, minimum range, in inches	1-3 (2.5-7.6 cm.)	3-4 (7.6-10.2 cm.)	4-6 (10.2-15.2 cm.)	1-3 (2.5-7.6 cm.)
Spindle speed, rpm minimum	8000	5500	3500	8000
Drive motor horsepower, minimum under full load at rated revolutions per minute.	1/3	1/3	1/3	1/3
Rated torque range, foot-pounds	0.3 to 0.5	0.3 to 0.5	0.3 to 0.5	0.3 to 0.5

3.6.1.2 Handle. The driver shall be fitted with a pistol grip, planer, "D", or straight-type handle affording the operator a full hand grip. The handle shall permit the operator (one man) to position and control the driver during the grinding operation.

3.6.1.3 Drive motor. The drive motor shall be of the universal type forming to CC-M-1807, Grade B insulation or better, and minimum horsepower as specified in Table I. Unless otherwise specified (see 6.2), the motor shall operate from a power source of 115 volts, 60 Hz, single phase and 115 volts direct current.

3.6.1.4 Switch. The motor shall be controlled by a trigger or button-type switch of quick-break design. Current rating of the switch shall not be less than the full load current rating of the motor. The switch shall not be equipped with a mechanism for locking in the "ON" position. Switches shall conform to the requirements of UL 45.

3.6.1.5 Power cord. The driver shall be furnished with not less than 15 feet of flexible, oil resistant, rubber-covered, 3-conductor cord, type SO or SJO in accordance with UL 73 for motor-operated appliances. For Navy procurement, the driver shall be furnished with not less than 15 feet of cable in accordance with type I requirements of J-C-175. The ground conductor shall have a green covering and shall be securely connected to provide a common ground with the exposed metal parts of the driver frame. The ground connection or connections shall be made on the inside of the

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driver frame. The size of the conductors shall be determined from the full load ampere rating of the motor and shall conform to NFPA 70. A 3-wire grounding plug shall be provided conforming to W-C-596/11 and have a cord grip.

3.6.1.6 Bearings. Bearings shall be ball or roller type of suitable design, housed to prevent leakage of lubricant and entrance of dirt and dust. The bearings shall be lubricated sealed-for-life type or shall be provided with means for proper lubrication application.

3.6.2 Pilots. The pilot sizes shall be in fractions of an inch as specified by the procuring activity (see 6.2). The pilots shall be of tool steel, heat treated to a hardness of 45 to 60 on the Rockwell C scale at any point along the length of the pilot which serves as the guide for the grinding wheel holder specified in 3.6.3. The difference in the diameter (taper) along the grinding wheel holder guide length of the pilot shall not exceed 0.0005 inch. The concentricity of any section along the wheel-holder guide length of the pilot shall not exceed 0.0005 inch tir. The accuracy of the pilot self-centering method shall permit a pilot, inserted and secured in an engine valve guide or valve guide cylinder block hole of corresponding size, to be relocated by turning the pilot 180 degrees to 190 degrees from its former position without altering its position by more than 0.001 inch when measured relative to fixed reference points on the valve seat. If expansion pilots are furnished, the minimum fit range shall be 0.002 inch below to 0.008 above the nominal diameters of the valve guide holes for the engines specified. Expanding pilots shall be stamped with either the nominal fractional size or within the range of expansion. If special hand tools are required for installing the pilots, these tools shall be furnished. The installation of the pilots shall not require the use of power tools. Unless otherwise specified (see 6.2), the quantity shown for the following expanding type pilots for type I, size 1, grinding kit shall be furnished:

1 each 1/4 inch
1 each 9/32 inch
2 each 5/16 inch
1 each 11/32 inch
2 each 3/8 inch
1 each 13/32 inch
1 each 7/16 inch
1 each 1/2 inch
1 each 9/16 inch
1 each 5/8 inch

Unless otherwise specified (see 6.2), two each of the following expanding type pilots for type II, size 1, grinding kit shall be furnished:

7/16 inch
15/32 inch
1/2 inch
9/16 inch

3.6.3 Wheel holder. Precision made wheel holders shall be ball or roller bearing type capable of retaining the grinding wheels specified in 3.6.4. If not an integral part of the spindle, the wheel holders shall incorporate provisions for connecting to the spindle drive in 3.6.1.1 without the use of tools. Unless otherwise specified (see 6.2), one wheel holder with threaded spindle shall be provided, having 11/16-16NF-2 or 9/16-16NF-2 threads. The threaded length of the spindle shall be not

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less than 9/16 inch. For Navy procurement, the wheel holder shall incorporate a threaded spindle embodying 9/16-16NS-2, 11/16-16NS-2, 7/8-14NF-2, or 1-1/8NEF-2 threads as specified by the procuring activity (see 6.2).

3.6.4 Grinding wheels. Unless otherwise specified (see 6.2), the quantity, size, seat angles, and threaded holes of the grinding wheels shall be as specified below. Wheels suitable for rough and finish grinding of both hard (stellite, etc.) and soft (cast iron, etc.) valve seats shall be furnished. The center hole of the grinding wheels shall be fitted with a metal insert incorporating female threads, providing a threaded hole for engaging the threaded spindle of the wheel holder (see 3.6.3). The grinding wheels shall have a minimum thickness of 3/4 inch. The grinding wheels shall embody the following tolerance requirements prior to any dressing operations:

a. The diameter of the grinding wheels shall be within a tolerance of $\pm$ 0.06 inch.

b. The thickness of the grinding wheels shall be within $\pm$ 0.0312 inch.

c. The concentricity of the grinding wheels shall be within a tir of 0.002 inch.

d. The run-out of the grinding wheel face (grinding surface) shall be within a tir of 0.002 inch.

e. The grinding wheels shall be properly balanced to preclude vibration during use.

f. The specified angle of the grinding wheel face shall be within $\pm$ 1 degree.

g. Grinding wheels, as specified above, shall be of the following quantities and sizes:

One each of the multipurpose grinding wheels listed below shall be contained and secured in the carrying case for the type I grinding kit. In addition, one each of the multipurpose grinding wheels listed below shall be supplied for the type I grinding kit as extras and packaged in accordance with 5.1.

1 1/2 inch diameter, 15 degrees bevel
1 5/8 inch diameter, 15 degrees bevel
1 3/4 inch diameter, 15 degrees bevel
2 inch diameter, 15 degrees bevel
2 1/4 inch diameter, 15 degrees bevel
2 1/2 inch diameter, 15 degrees bevel
2 3/4 inch diameter, 15 degrees bevel
3 inch diameter, 15 degrees bevel

1 1/2 inch diameter, 30 degrees bevel
1 5/8 inch diameter, 30 degrees bevel
1 3/4 inch diameter, 30 degrees bevel
2 inch diameter, 30 degrees bevel
2 1/4 inch diameter, 30 degrees bevel
2 1/2 inch diameter, 30 degrees bevel
3 inch diameter, 30 degrees bevel

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1 1/4 inch diameter, 45 degrees bevel
1 3/8 inch diameter, 45 degrees bevel
1 1/2 inch diameter, 45 degrees bevel
1 5/8 inch diameter, 45 degrees bevel
1 3/4 inch diameter, 45 degrees bevel
2 inch diameter, 45 degrees bevel
2 1/4 inch diameter, 45 degrees bevel
2 1/2 inch diameter, 45 degrees bevel
2 3/4 inch diameter, 45 degrees bevel
3 inch diameter, 45 degrees bevel

Two each of the multipurpose grinding wheels listed below shall be contained and secured in the carrying case for the type II grinding kit.

2 1/2 inch diameter, 30 degrees bevel
3 inch diameter, 30 degrees bevel
2 1/4 inch diameter, 45 degrees bevel
2 1/2 inch diameter, 45 degrees bevel
3 inch diameter, 45 degrees bevel

For Navy procurement, the grinding wheel holes shall be fitted with a brass or steel insert incorporating female threads as specified by the procuring activity (see 6.2).

3.6.4.1 Grinding wheel marking. The grinding wheels shall be marked with diameter, wheel bevel angle, manufacturer's name or trademark, manufacturer's number, and maximum operating speed recommended. If the grinding wheels do not permit such marking, the container they are furnished in shall be permanently identified as cited above.

3.6.5 Dressing stand. The dressing stand shall incorporate facilities to dress (refinish) the grinding surfaces of the grinding wheels in 3.6.4 when the grinding wheels are mounted to the wheel holder and driven by the spindle drive motor. The dressing stand shall be capable of dressing the grinding surface of the grinding wheels within a concentricity of 0.001 inch tir reading, and within + 10 minutes for any indicated angle. The accuracy of the dressed grinding wheel shall be measured from the axis of the wheel holder, in its dressed condition without disassembly of the grinding wheel from the wheel holder. The dressing stand shall consist of the following principal components.

3.6.5.1 Base. The stand base shall maintain the dressing stand in a stable position without movement or vibration during the grinding wheel dressing operation.

3.6.5.2 Stand pilot. The stand pilot shall position the grinding wheel workhead to effect the grinding wheel dressing operation by the wheel dressing unit in 3.6.5.3. The pilot, if adjustable, shall be capable of being positioned without the use of tools.

3.6.5.3 Wheel dressing unit. The wheel dressing unit shall embody facilities to manually traverse a dressing tool (diamond) across the grinding face of the grinding wheels at a specified angle. The wheel dressing unit shall include an industrial diamond dressing tool mounted on a holder with a feed device and a quadrant. The dressing tool shall be removable and adjustable manually without the use of tools. The quadrant shall have a marked angle range of 0 degrees to 90 degrees. The quadrant shall have increments and numerical markings at least every 15 degrees to provide indicated angle settings of 15 degrees, 30 degrees, and 45 degrees. The dressing tool holder shall be capable of being securely positioned at any angle within the quadrant

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range to permit dressing the grinding face of the grinding wheels at the set angle. The setting adjustment shall be accomplished without the use of tools.

3.6.6 Valve seat indicator. When specified (see 6.2), the required valve seat indicator shall essentially consist of a dial indicator gage with a follower and mounting to permit utilizing the gage with the pilots specified in 3.6.2. The indicator dial shall be not less than 1-1/2 inches in diameter and have an indicating range of not less than 0 to 0.050 inch. The indicating range shall be divided into increments of not more than 0.001 inch with numerical markings at least every 0.010 inch. The error in the dial reading shall not exceed 0.0005 inch. The dial markings and dial background shall be of contrasting colors. The indicator shall provide a total reading reflecting the accuracy of the finished ground valve seat relative to the center line of the valve guide hole within the capacities for the types and sizes specified in Table I.

3.6.7 Carrying case. Each grinding kit shall incorporate a carrying case fabricated from sheet metal not less than Manufacturer's Standard Gage No. 20. The case shall be fitted with a hinged cover provided with two positive acting latches. The case shall incorporate one or two folding handles to permit a full-hand grip for carrying the case with components enclosed. The carrying case shall provide protection, secure retention, and support for the following grinding kit components: grinding wheels, pilots, wheel holders, dressing stand, valve seat indicator, and the driver. The components shall be easily removed and replaced so there is no interference with other components. The carrying case shall provide necessary protection for all components to withstand without damage the vibration and shock requirements specified herein.

3.7 Electromagnetic compatibility. When specified (see 6.2), the grinding kits shall be in compliance with the requirements of MIL-STD-1337.

3.8 Dielectric strength. The electric-powered driver of the grinding kit shall withstand without damage, a root mean square (rms) voltage of 900 plus twice its rated voltage, applied for a period of 1 minute between mutually insulated circuits and between insulated circuits and ground; the voltage being at a frequency of 60 Hz.

3.9 Fungus control. Components of electrical circuitry that are hermetically sealed or inherently resistant to fungus and moisture need not be treated. When specified (see 6.2), components of electrical circuitry that are not inherently resistant to fungus growth or moisture shall be treated in accordance with MIL-T-152 and MIL-V-173.

3.10 Accessory equipment. The grinding kits shall include all required tools, except common hand tools, for operating the grinding kit.

3.11 Painting. Unless otherwise specified (see 6.2), painting and finishing shall be in accordance with the best commercial practice in the industry. Before painting all surfaces to be painted shall be clean and be free of all foreign matter detrimental to painting and at least one coat of primer (or a phosphate or chromate base) and one coat of enamel, or equivalent (i.e., epoxy enamel) is applied.

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3.12 Product identification. The grinding kits shall be marked for product identification in accordance with MIL-STD-130 and, unless otherwise specified (see 6.2), shall include the National Stock Number.

3.13 Workmanship. Workmanship of the grinding kits and accessories shall be of the best quality prevailing among manufacturers of this type of equipment. The items shall be free from all imperfections which will adversely affect the general appearance, function, and serviceability.

4 QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspection. The inspection requirements specified herein are classifed as follows:

a. First article inspection (see 4.3).

b. Quality conformance inspection (see 4.4).

4.3 First article inspection. First article inspection shall be applied to the preproduction model or initial production items (see 3.1). Unless otherwise specified (see 6.2), first article inspection shall consist of the examination in 4.5 and all tests under 4.6. Failure of the first article to pass the examination or any of the tests shall be cause for rejection.

4.4 Quality conformance inspection. Unless otherwise specified (see 6.2), quality conformance inspection shall be applied to production items offered for acceptance under the contract. Unless otherwise specified (see 6.2), quality conformance inspection shall consist of "a" through "f" as follows. Failure of any unit to pass an examination or test shall be cause for rejection of the item.

a. Product examination (see 4.5).

b. Performance test (see 4.6.1).

c. Accuracy and tolerance tests (see 4.6.2).

d. Vibration test and shock test (see 4.6.3 and 4.6.4).

e. Electromagnetic and dielectric tests (see 4.6.5 and 4.6.6).

f. Packaging inspection (see 4.7).

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4.4.1 Sampling. Unless otherwise specified (see 6.2), sampling for quality conformance inspection shall be as specified in MIL-STD-105 using the following:

Inspection or Test	Inspection Level	AQL
a. Product examination (see 4.5).	II	1.0
b. Performance test (see 4.6.1).	II	1.0
c. Accuracy and tolerance tests (see 4.6.2).	II	1.0
d. Vibration and shock tests (see 4.6.3 and 4.6.4).	S-4	1.0
e. Electromagnetics and dielectric (see 4.6.5 and 4.6.6).	S-4	4.0
f. Packaging inspection (see 4.7).	S-4	4.0

Nonconformances shall be expressed in terms of Defects Per Hundred Units.

4.5 Product examination. Visually and manually examine the grinding kit to determine conformance with the requirements of section 3. Visual examinations shall include verification of completeness of manufacture and assembly, conformance to specified standards, adequacy of markings, proper cleaning, and freedom from the identified defects. Manual examinations shall include the operation of movable parts by hand to assure proper functioning. The examination provisions may be applied at the earliest practical point in manufacture at which it is feasible to inspect for acceptance without risk of change in the characteristic by subsequent operations. Failure of the contractor to provide objective evidence that the grinding kit passed the examinations prescribed for them by the contractor's inspection system shall be cause for rejection.

4.6 Test methods.

4.6.1 Performance test. The applicable used engines or test mock-up employing valves within the capacity range as specified in Table I shall be utilized. The valve guides shall be cleaned and tested to be within the tolerance as specified by the engine manufacturer prior to the valve seat grinding operation. At least four intake and four exhaust valve seats shall be ground in accordance with the grinding kit manufacturer's instructions. During and on conclusion of the valve seat grinding operation the grinding kit characteristics shall be observed for compliance with the following:

a. Functional requirements as specified in 3.5.1.1 for type I kits and in 3.5.1.2 for type II kits.

b. Spindle drive function and operation as specified in 3.6.1.1.

c. Positioning and control of the driver as specified in 3.6.1.2.

d. Driver switch operation as specified in 3.6.1.4.

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e. Pilot installation in valve guide or valve guide hole as specified in 3.6.2.

f. Wheel holder connection to spindle drive and to grinding wheel as specified in 3.6.3.

g. Dressing stand function and operation as specified in 3.6.5 and 3.6.5.3.

h. The finish ground valve seat shall be tested for accuracy utilizing the valve seat indicator specified in 3.6.6, when furnished, or standard measuring equipment with an accuracy of 0.0001 (after the applicable pilots and the valve seat indicator have been checked in accordance with 4.6.2.1 and 4.6.2.4) to determine compliance with 3.5.2.

4.6.2 Accuracy and tolerance tests.

4.6.2.1 Pilot tolerance tests. The pilots shall be subjected to the following tests.

4.6.2.1.1 Taper test. Utilizing a measuring method having an accuracy within 0.0005 inch, the diameter of the wheel holder guide length portion of the pilots shall be measured along its complete length. The difference in the diameter readings shall be not more than 0.0005 inch to comply with 3.6.2.

4.6.2.1.2 Concentricity test. Utilizing a measuring method having an accuracy within 0.0005 inch, the concentricity of the diameter of the wheel holder guide length shall be tested at not less than five points along the guide length portion of the pilot. The concentricity of each section tested shall not be altered by more than 0.0005 inch tir to comply with 3.6.2.

4.6.2.1.3 Straightness test. With a mock-up utilizing the applicable new valve guide or gage, the pilot shall be secured in position in the valve guide and its position located by a measurement from a fixed reference point. The pilot shall be loosened and relocated by turning it 180 degrees to 190 degrees from its former position. The measurement from the fixed reference point to the pilot shall not be altered by more than 0.001 inch to comply with 3.6.2.

4.6.2.2 Grinding wheel tolerance tests. The grinding wheels, prior to any dressing operation on the wheel dressing stand, shall be tested as specified below.

4.6.2.2.1 Wheel size test. Utilizing a measuring device within a accuracy of 0.001 inch, the specified diameter and thickness of the grinding wheels shall comply with 3.6.4.

4.6.2.2.2 Concentricity test. Utilizing a dial indicator having an accuracy within 0.001 inch, the concentricity of the grinding wheels shall be tested about their outside diameter. The total indicator reading shall be observed and shall be in compliance with 3.6.4.

4.6.2.2.3 Grinding surface angle test. The angle of the grinding wheel grinding surface shall be measured by a method having an angle accuracy within 15 minutes. The angle of the grinding surface shall be measured at least at four points about the grinding surface circumference and the measurements shall be in compliance with the angle tolerance requirements specified in 3.6.4.

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4.6.2.3 Dressing stand accuracy test. The grinding wheels shall be dressed in the dressing stand to the specified angle. Upon completion of the wheel dressing operation, the grinding wheels shall be tested for concentricity and angle accuracy as stated below:

a. Utilizing a dial indicator having an accuracy within 0.0001 inch, the concentricity of the grinding wheels shall be tested about the dressed grinding surface. The total indicator reading shall be observed to determine compliance with 3.6.5.

b. The angle of the dressed grinding surface shall be measured at least at four points about the wheel grinding surface, employing a method having an accuracy of within 1 minute for any angle in its measurement range. The angle readings shall be observed to determine compliance with 3.6.5.

4.6.2.4 Valve seat indicator accuracy test. If a valve seat indicator is furnished (see 3.6.6), it shall be tested in a mock-up utilizing an applicable new valve guide or a gage. The measuring accuracy of the valve seat indicator using the standard applicable pilot shall be tested against a measuring method having an accuracy within 0.0001 inch. The accuracy of the valve seat indicator and pilot method shall be in compliance with 3.6.6.

4.6.3 Vibration test. The grinding kit shall be subjected to vibration in compliance with 3.5.3. At the conclusion of the test, the grinding kit shall be examined for evidence of damage and tested in accordance with 4.6.1 and 4.6.2 to determine compliance with 3.5.2.

4.6.4 Shock test. The grinding kits shall be subjected to shock in compliance with 3.5.4. At the conclusion of the test, the grinding kits shall be examined for evidence of damage and tested in accordance with 4.6.1 and 4.6.2 to determine compliance with 3.5.2.

4.6.5 Electromagnetic compatibility. When required, the grinding kits shall be tested in accordance with the requirements of MIL-STD-1337 (see 3.7).

4.6.6 Dielectric test. The electric-powered driver of the grinding kit shall be subjected to twice rated voltage plus 900 volts rms at 60 Hz, applied for a period of 1 minute. The driver shall then be inspected for evidence of damage to determine compliance with 3.8.

4.7 Packaging inspection. Packaging inspection shall be conducted before and after packaging to determine compliance with the requirements of Section 5.

5 PREPARATION FOR DELIVERY

5.1 Preservation-packaging and packing.

5.1.1 Department of Defense.

5.1.1.1 Military preservation-packaging and packing. When specified (see 6.2), level A preservation-packaging and levels A or B packing shall be in accordance with PPP-P-40. The additional grinding wheels which are furnished with type I grinding kit shall be packaged in accordance with method III criterion of MIL-P-116. Place the packaged wheels inside the unit container which houses the kit.

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5.1.1.2 Industrial preservation-packaging and packing. When specified (see 6.2), industrial preservation-packaging and packing shall be in accordance with ASTM D3951-82.

5.1.2 Civil agencies. When specified (see 6.2), commercial preservation-packaging and packing shall be that normally used by the supplier for retail distribution, shall afford adequate protection against physical and environmental damage during shipment from the supplier to the first receiving activity, and shall comply with the rules and regulations applicable to the mode of transportation.

5.2 Marking.

5.2.1 Military marking. In addition to any special marking required by the contract or order, interior packages and exterior shipping containers shall be marked in accordance with the requirements of MIL-STD-129.

5.2.2 Industrial marking. Industrial marking shall be in accordance with ASTM D 3951-82.

5.2.3 Civil agencies marking. In addition to any special marking required by the contract or order, interior packages and exterior shipping containers shall be marked in accordance with the requirements of FED-STD-123.

6 NOTES

6.1 Intended use. The grinding kits covered by this specification are intended for use in grinding intake and exhaust valve seats on both automotive and aircraft engines.

6.2 Ordering data. Purchasers should select the preferred options permitted herein and include the following information in procurement documents:

a. Title, number, and date of this specification.

b. Type and size required (see 1.2.1).

c. First article, when required (see 3.1).

d. Special safety devices, when required (see 3.4.8).

e. Voltage, if different (see 3.6.1.3).

f. Pilot sizes required, if different (see 3.6.2).

g. Quantity of pilots, if different (see 3.6.2).

h. Wheel holder thread size, if different (see 3.6.3).

i. Quantity, sizes, and seat angles of grinding wheels required (see 3.6.4).

j. Valve seat indicator, when required (see 3.6.6).

k. Electromagnetic compatibility, when required (see 3.7).

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l. Fungus control of electrical components, circuit elements, and fungus control of electrical motors, when required (see 3.9).

m. Painting, if different (see 3.11).

n. Identification marking, if different (see 3.12).

o. First article inspection, if different (see 4.3).

p. Quality conformance inspection, if different (see 4.4).

q. Sampling, if different (see 4.4.1).

r. Preservation-packaging and packing, if different (see 5.1).

s. Selection of applicable level of packaging and packing (see 5.1).

6.3 Contract data requirements (for military acquisition). Required technical data such as operator's manuals, parts lists, wiring diagrams, and other instructions for operation and maintenance, as identified on a numbered DD Form 1664, should be specified on DD Form 1423 incorporated into the contract.

6.4 Safety and health requirements. In order that equipment integrated into the user's operational environment will comply with OSHA limitations and control of noise levels, radiation, electromagnetic emission, noxious vapors, heat, etc., as applicable, specific requirements concerning such points of operation, and other health and safety requirements, should be specified by the user.

6.5 The values stated in US customary units are to be regarded as the standard. The metric equivalents of US customary units given in this specification may be approximate.

MILITARY INTEREST:

Custodians

Army - AL
Navy - YD
Air Force - 99

User activities

Army - AT, ME

CIVIL AGENCY COORDINATING ACTIVITIES:

GSA-FSS

PREPARING ACTIVITY:

ARMY - AL

DoD Project 4910-0564

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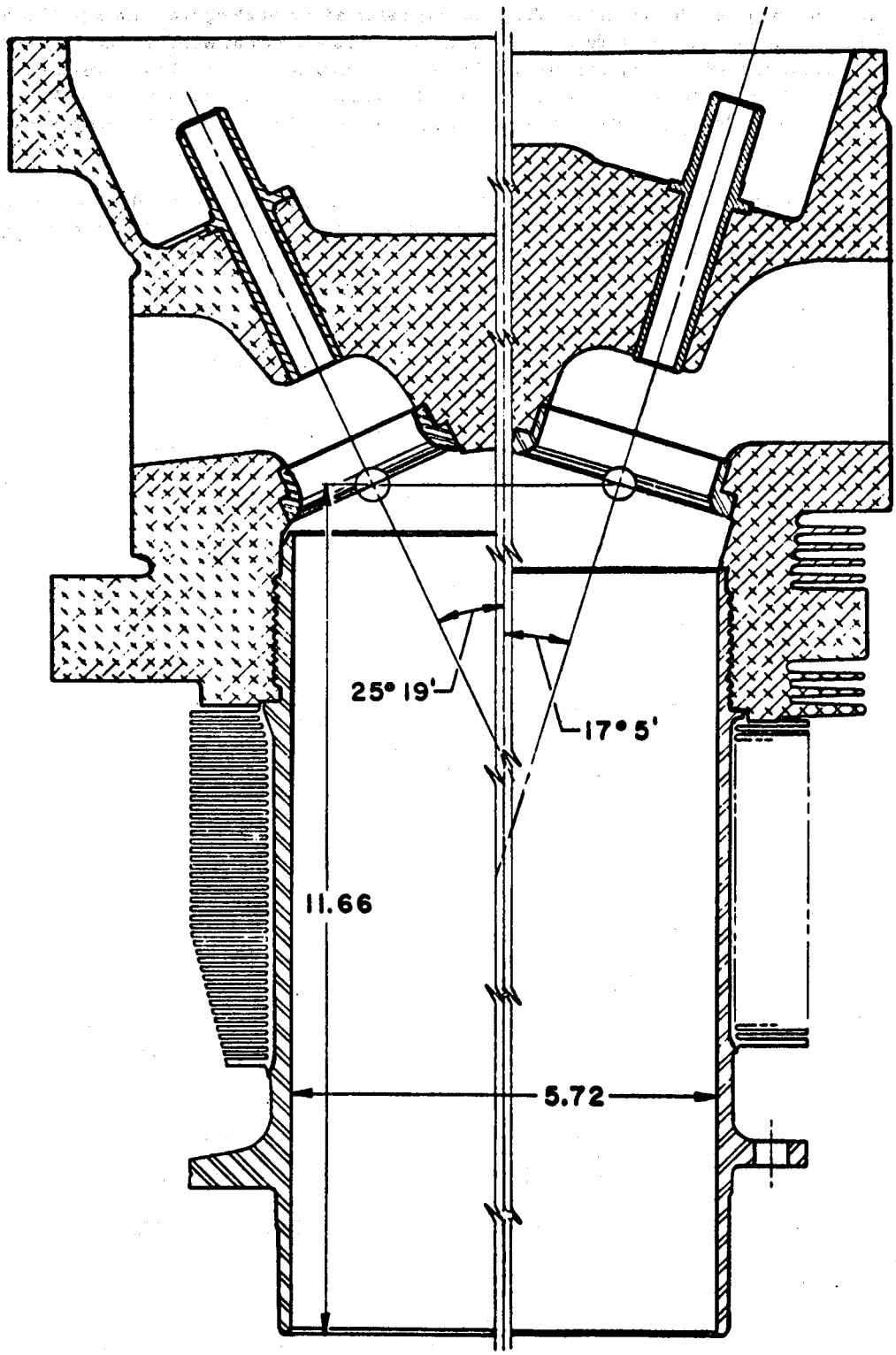


FIGURE 1 - ENGINE CYLINDER CONFIGURATION

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